When Metrics Say "Yes" But Humans Say "No"

A Case Study of Hallucination Reduction for Factual Summarization

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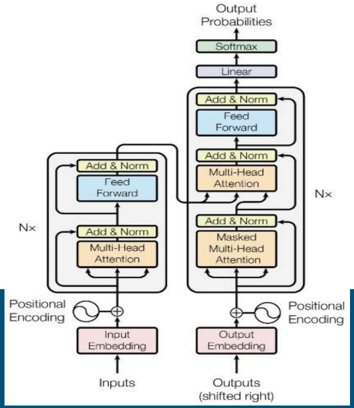
Project Characters

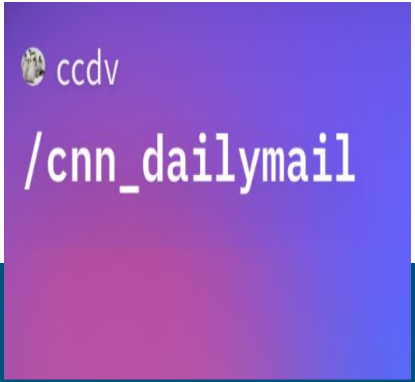


BART(Generator)

Fine-tuned summarization model

FactCC(Verifier)

Classifier that scores article summary factual consistency.



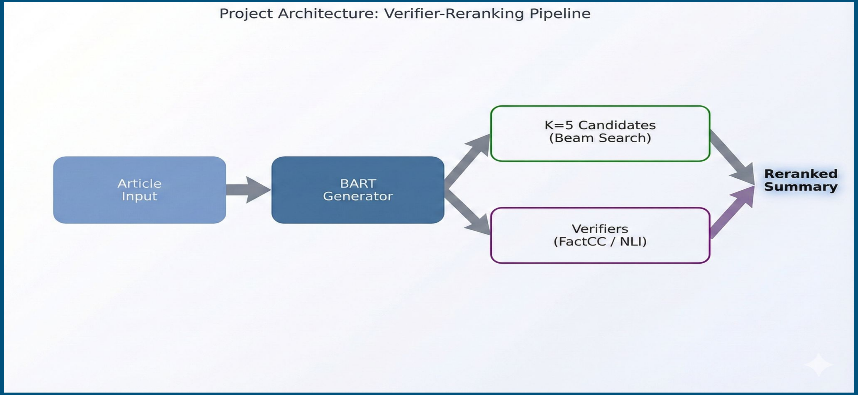
CNN/Daily Mail (Dataset)

News articles with highlight summaries

Single Human Annotator

Single rater doing a 50-example audit.

Task/Problem/Goal



Test whether better automatic metrics ⇒ better human-judged summaries.

Success Criteria: ΔFactCC ≥ +2.0, ROUGE-L drop ≤ 1.0

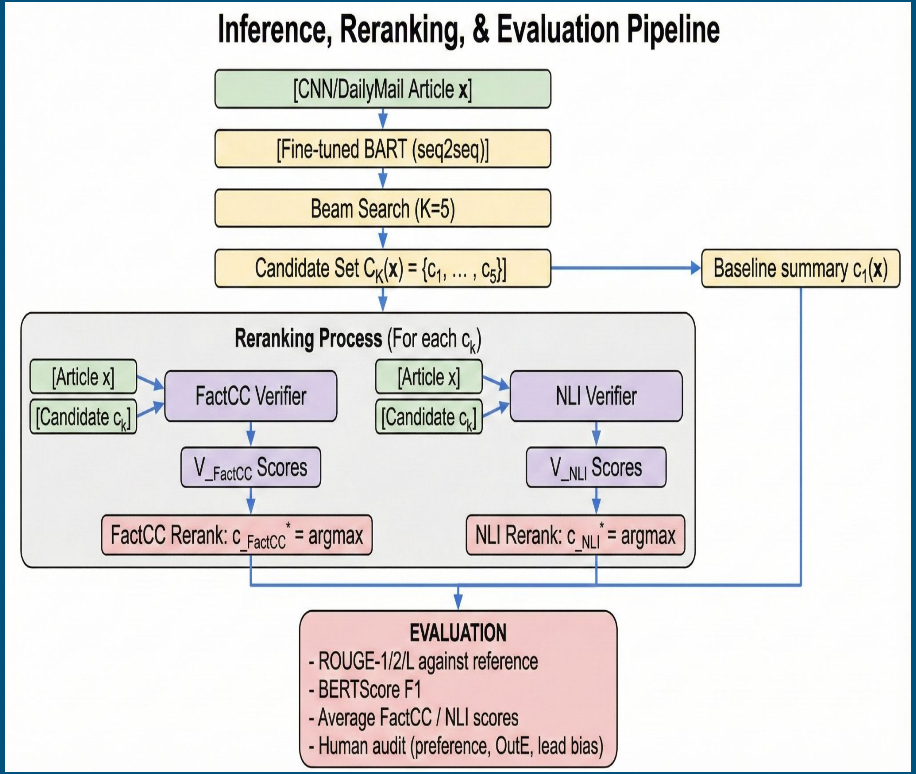
Task:
CNN/DailyMail abstractive summarization (BART-base)

Problem:
Summaries are fluent but sometimes factually wrong (hallucinations)

Goal:
Improve factual consistency **without** retraining a bigger model, use reranking instead.

Architecture Diagram (Generate → Verify)

- Generate K=5 candidates with BART (beam search)
- Score each candidate with:
  - FactCC: probability the summary is consistent with the article.
  - RoBERTa-MNLI: entailment score (for ablation).
- Pick the candidate with highest FactCC score.



BART + FactCC / Experimental Setup

Models

- **Generator: facebook/bart-base (fine-tuned on 20k CNN/DM, 3 epochs, lr=5e-5) -**
  a. **Verifier A: FactCC (factual consistency) -**
  b. **Verifier B: RoBERTa-MNLI (ablation)**

Data

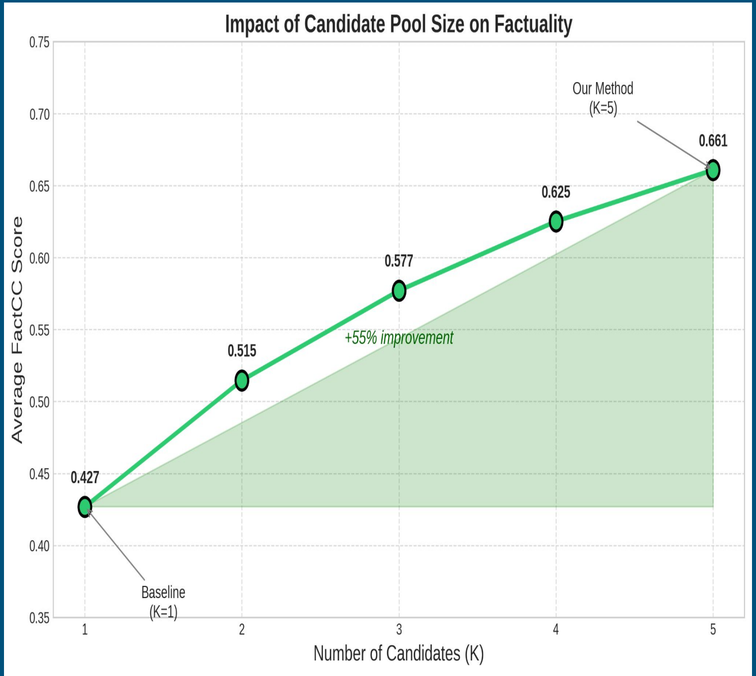
- **Train: 20k subset (seed=42) - Val: 2k | Test: 11,490**

Decoding

- **Beam search, K=5, max length 140**

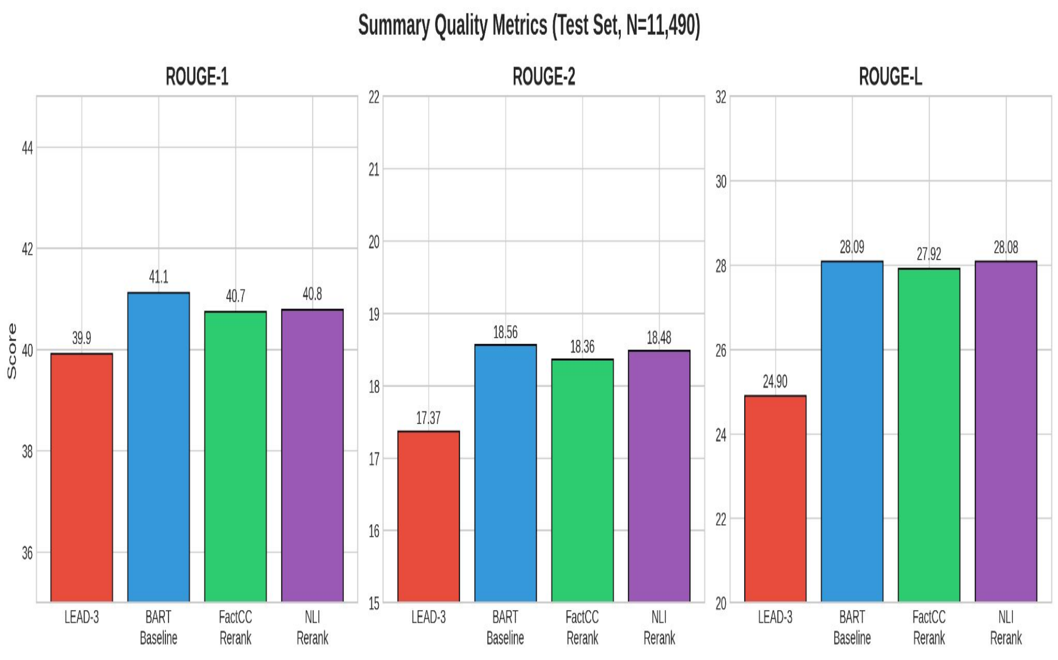
Success Criteria ✓ ΔFactCC ≥ +2.0 points ✓ ROUGE-L drop ≤ 1.0 point

After analyzing K-ablation shows strong gains up to ~3–5, with diminishing returns.



Automatic Results

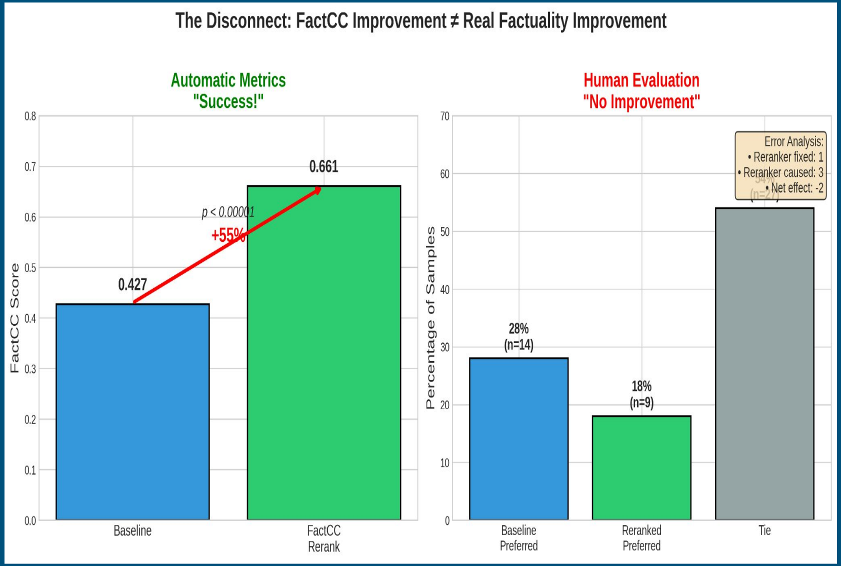
- ROUGE-L:
  - LEAD-3: 24.9
  - BART baseline: 28.1
  - FactCC rerank: 27.9
  - NLI rerank: 28.1
- FactCC (avg factuality score):
  - Baseline: 0.427
  - FactCC rerank: 0.661 (+55%, $p < 0.00001$)
- Success criteria satisfied:
  - ΔFactCC ≥ +2.0 ✅
  - ROUGE-L drop ≤ 1.0 (−0.17) ✅



Setting / *N = 50* human audit

. `df[df['changed']==True].sample(30)` and `df[df['changed']==False].sample(20)`

For this project, a ***hallucination*** is any statement in the summary that is not supported by, or contradicts, the source article, including entity, event, temporal, and out-of-thin-air errors.



- 50 test examples:
  - 30 where baseline vs rerank disagree
  - 20 where they agree
- Human (Single Annotator) preference:
  - Tie: 27 (54%)
  - Baseline preferred: 14 (28%)
  - Reranked preferred: 9 (18%)
- Net hallucination effect: reranker fixes 1 error, introduces 3 → ***Net −2***

When Reranking Helps vs Hurts

The Success (Logic Repair)

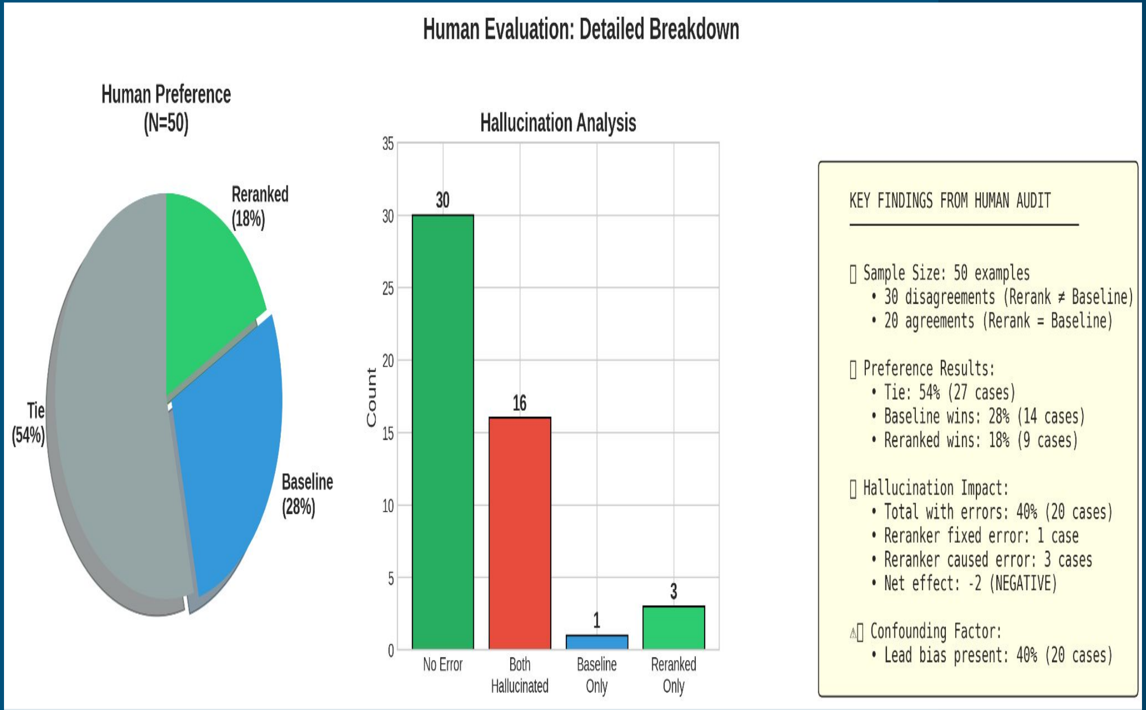
- **Context:** Refugees formed a human chain to stop attackers.
- **Baseline (The Mistake):** *"15 men arrested... But were **prevented** [from being arrested] because they huddled together..."*
    - *Status:* ***Logical Error*** (Implies the chain stopped the police).
- **Reranked (The Fix):** *"...claimed the men tried to throw other Christians off... but were **prevented** [from doing so] because they huddled together..."*
    - *Status:* **Corrected** (The verifier caught the causal mismatch).

The Failure (Hallucination)

- **Context:** A father hit his toddler; store owner gave video to police.
- **Baseline (The Safe Bet):** *"Justin Whittington... was taken into custody... relationship not yet revealed..."*
    - *Status:* ***Boring but Factual*** (Stuck to the text).
- **Reranked (The Lie):** *"...Police were notified... and **asked the public for help identifying the man.**"*
    - *Status:* **Hallucination** (The article never mentioned a public appeal).

Verifier (FactCC) isn't actually checking facts it's just checking for patterns.

Conclusion



- **Lead bias**: CNN/DailyMail favors early sentences; LEAD-3 gets ROUGE-L 24.91 (only 3.18 points below BART). Reranking often moves toward extractive “lead-like” summaries without improving factuality.

- **FactCC mis-scores paraphrases**: Penalizes legitimate paraphrasing, prefers extractive but slightly wrong spans.

- **Limited beam diversity**: in ~32% of audited examples all K=5 candidates had hallucinations → reranking cannot fix generator-level failures.

- **Why:** Goodhart's Law. Once I made FactCC the optimization target, the system learned to game FactCC rather than be more truthful.

References & Github

- Lewis et al. 2020. “BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation.” *ACL.*
- Hermann et al. 2015; Nallapati et al. 2016. CNN/DailyMail summarization dataset.
- Kryscinski et al. 2020. “Evaluating the Factual Consistency of Abstractive Text Summarization.” (FactCC).
- Ji et al. 2023. “Survey of Hallucination in Natural Language Generation.”
- Zha et al. 2023. “How Well Do Factuality Metrics Align with Human Judgments?” (for your QAGS/FactCC alignment discussion).

https://github.com/rPowDS/MIDS266-NLP-final-project